

Ease Your Numerical Analysis With OCTAVE 4.2.1

To calculate the area of a circle, you can follow the simple formula of πr^2 (where 'r' is the radius of the circle). But, calculating the mathematical value of ' π ' beyond nine digits involves a technique called numerical analysis. When designing complex products such as aircrafts or rockets, Octave 4.2.1 can be a boon for designers.

Octave is a special high-level software used for numerical computation. It is basically a solution provider to linear and non-linear numerical problems.

Scientific Octave and how it works

Octave is a powerful scientific programming language that runs on GNU/Linux, MacOS, BSD and Windows. Users can solve all the linear/non-linear algebra and statistical analysis and other numerical experiments using Octave.

On most systems, Octave runs with the shell command 'octave,' where it displays a graphical user interface.

Octave can do elementary calculations, create a matrix, use integral differential equations, produce a graphical output, edit and document all the processes. Each of these calculations involves a set of programs. For example, to verify Euler's identity, you need to follow the simple steps given below:

$$e^{i \times \pi} = -1$$

octave:1 > exp (i x pi)

Besides, Octave has a help facility where documents in printed form are also available in Octave prompt as creation files for both

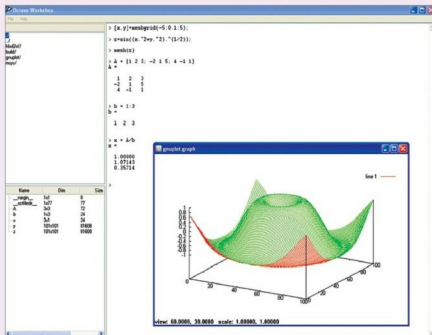


Fig. 1: Octave work plot (Image courtesy: <http://ccm.net/download>)

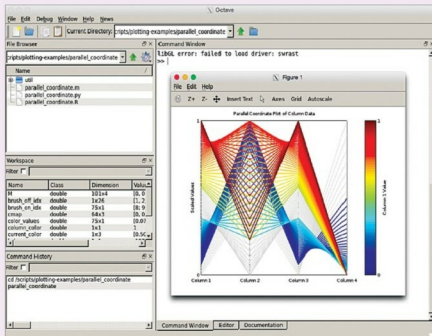


Fig. 2: Octave plotting snapshot (Image courtesy: <http://blog.ctaggart.com>)